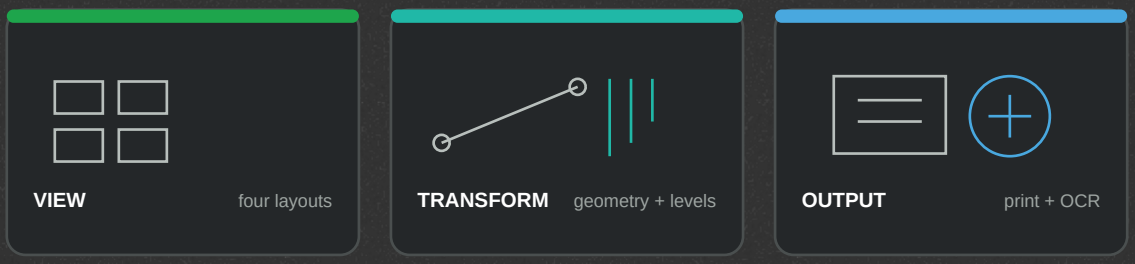
KillerPDF

Technical Specification & Complete Feature Brochure

VERSION 1.8



A native Windows PDF viewer, annotator and light editor.
Powered by The KillerPDF.Engine. Built on .NET 10, WPF and PDFium.

This document is itself a KillerPDF showcase - open it in the app
and explore the outline, forms, themes, and annotation pages.

KillerPDF.net

.NET 10 - x64 - Windows 10/11 - GPL-3.0

About the Maker

KillerPDF

Steve the Killer

Field technician - systems administrator - builder of Killer Tools

KillerPDF was born out of a ticket queue.

I spend my days as a field tech and systems administrator at a large MSP, and a steady share of that work is always the same shape: a user can't fill a PDF, a form won't flatten, a license won't activate, a viewer throws an error on a file that should just *open*. Adobe Acrobat tickets - install it, repair it, license it, work around it - added up to hours every week that never felt like they were actually fixing anything.

So I built the tool I wished my users had: one fast, self-contained executable that opens the broken files, fills forms, signs documents, OCRs scans and gets out of the way. No account, no subscription, and installation is optional. overbuilt for the encrypted, broken-xref and rotated-and-clipped PDFs that make other viewers give up - because those are the files that generate tickets. Everything in this brochure is real and shipping.

> "I got tired of closing the same Adobe ticket. So I wrote a PDF app that just works, ships as one .exe, and never asks the user to log in. That app is KillerPDF."

KillerPDF is free software (GPL-3.0). Find it, and everything else I build, at **KillerPDF.net**.

About this Brochure

This document is the complete technical specification and feature list for KillerPDF version **1.8**, and a working showcase of the app itself.

Open it inside KillerPDF and it becomes a tour: the outline is a deep nested bookmark tree, Part IV has a real interactive AcroForm and a printed-style form to mark up, and the annotation pages are seeded for screenshots. Every tool, theme, shortcut and constant here is taken directly from the source.

CURRENT 1.8 SERIES

The 1.8 series adds Ukrainian, Norwegian Bokmål, and Brazilian Portuguese interface translations alongside its earlier languages. Dark, Light, and Black now offer yellow and magenta as well as six existing accents.

The reference section follows the current shortcut map, theme choices, and language catalog. The validation tables retain their dated 1.8.3 measurements as historical results, not a claim about this development build.

HOW THIS DOCUMENT IS ORGANIZED

- **Part I** - platform, architecture, render pipelines, view modes, and calculation details.
- **Part II** - every tool in the GUI and under the hood, including the command-line interface.
- **Part III** - keyboard map, themes, languages, robustness, standards, and hard numbers.
- **Part IV** - a live showcase with interactive forms and annotation-ready pages.

Table of Contents

PART I | PLATFORM AND ARCHITECTURE

Product and tech stack 06
The KillerPDF.Engine 07
View modes 08
Rendering and zoom mathematics 09
Interaction mathematics 10

PART II | TOOLS AND FEATURES

Text, highlight, ink, shapes and images 12
Crop, links and forms 17
Stamp, search, OCR and signing 20
Print, transform, perspective, pages and import 24
Split pane, tabs, outline, redo, zoom and undo 28
Command line 32

PART III | REFERENCE

Keyboard shortcuts and tool keys 34
Themes, accents and localization 38
File handling and standards 40
Engine validation and guarantees 42
Specifications, constants and glossary 43

PART IV | LIVE SHOWCASE

Interactive and printed forms 46
Annotation playground 48
Colophon 50

Platform & Architecture

- The technology and ideas the app is built on
- Composed WPF shell, reusable document views
- A zoom-stable coordinate space shared by every view

KillerPDF

PLATFORM

Product & Tech Stack

KillerPDF is a native Windows PDF viewer, annotator, editor, OCR workstation, form tool, and document automation app. PDFium renders pages, PdfPig reads text, Tesseract performs OCR, and The KillerPDF.Engine owns document creation and writing.

Attribute	Value
Version	1.8 (mirrors the csproj version)
Target	.NET 10, WPF, WinExe, Windows x64
Distribution	Compact installer or self-contained portable package
Render / text	PDFium via Docnet.Core 2.6 / PdfPig 0.1.15
Write / signing	The KillerPDF.Engine / CMS SHA-256 digital signatures
Packaging	Verified payloads with per-user and all-users install protection
OCR	Tesseract 5.2, nineteen downloadable language models
Command line	Merge, split, decrypt, rasterize, flatten, print, OCR, and batch-resave

AT A GLANCE

- Four view modes, including Two-Page book layout, share one stable annotation coordinate space.
- Split view hosts two independent document panes; night-mode invert is stored and rendered per pane.
- Compare two PDFs with synchronized navigation, visual difference regions, and changed-page percentages.
- A full annotation, forms, OCR, signing, measurement, page editing, print, and Transform toolset.
- Thirteen themes, eight accent variants on base themes, and nineteen interface languages.

PLATFORM

The KillerPDF.Engine

Version 1.8.0 replaces the legacy write pipeline with an independent, UI-free .NET 10 document library. The desktop app consumes the same public API available to other applications. The engine has no dependency on WPF, KillerPDF application code, PDFium, PdfPig, PdfSharpCore, or PDFsharp.

A REAL LIBRARY BOUNDARY

Boundary	Responsibility
Desktop application	WPF interface, PDFium rendering, workflows, sessions, tools, and user interaction
The KillerPDF.Engine	PDF syntax, object graphs, validation, editing, writing, signing, encryption, and standards
Public API	Document creation, page import, content builders, forms, annotations, metadata, and save options
Test boundary	1,443 engine tests plus a reproducible 47,024-file corpus harness

CAPABILITIES

- Reads and validates cross-reference tables and streams, object streams, encryption, metadata, forms, annotations, outlines, page trees, and tagged structure.
- Authors PDF 2.0 and modern PDF/A, signs and encrypts documents, imports pages, edits existing object graphs, and supports incremental or deterministic full rewrites.
- Creates appearance streams for forms and annotations, including Unicode text, choice fields, checkboxes, radio buttons, comb fields, and signature widgets.
- Uses fail-closed validation before output, so unsupported or unsafe structures are rejected instead of silently producing a damaged file.

> Validation: 1,443 engine tests, 289 application tests, and a 47,024-input corpus. The complete release results appear on page 42.

View Modes

Four view modes share the same 2048-longest-side annotation space, so marks do not drift when the layout changes.

Choose Single, Continuous, Two-Page, or Grid using the four buttons in the bottom-right footer. The active mode is highlighted. Zoom out, the editable percentage and fit selector, and zoom in sit beside them. These controls act on the focused pane.

Mode	Behavior
Continuous (F5)	Vertical stream of every page; nearby bitmaps render asynchronously and distant ones leave the cache.
Single (F6)	One page in the primary tile with exact fit and zoom math for that page.
Two-Page (F7)	Facing-page spreads. Press B for book layout, which shows the cover alone and pairs the remaining pages like a physical book.
Grid (F8)	Whole-document thumbnail grid. Zoom changes the stored column count and the viewport determines the current page.

The moon button or N key inverts only the focused pane. The other pane can remain in its original colors, and focus immediately updates the rail icon. Shift+N controls whether images are included in inversion.

> F10 creates the second pane. Each pane keeps its own tabs, current page, zoom, view mode, sidebar position, selection, tool state, render cache, and night-mode invert state.

How It's Calculated: Rendering & Zoom

Two numbers describe every page: where things *are* (zoom-independent) and how many pixels we *draw* (zoom-dependent). Keeping them separate is what makes annotations stay put while text stays crisp.

1. The render-dim space

Every page is normalized so its longest side is exactly 2048 units; annotation coordinates are stored here, identical at every zoom and across all four pipelines.

```
maxDim = max(pageWidthPt, pageHeightPt)
rdW = round(2048 * pageWidthPt / maxDim)
rdH = round(2048 * pageHeightPt / maxDim) // longest side -> 2048
```

2. Decoupled bitmap resolution

The bitmap rasterizes at a resolution that follows DPI and zoom, so a 3x page is drawn from 3x the pixels rather than upscaled. The 6144 px ceiling bounds memory.

```
scaledMax = min( 6144, int( 2048 * max(dpiScaleX, dpiScaleY) * max(1.0, zoom) ) )
```

3. The coordinate Y-flip

PDF uses a bottom-left origin in points; WPF canvases a top-left origin in DIP. Every mark crosses that boundary:

```
sx = renderW / pdfW sy = renderH / pdfH
canvasX = left * sx
canvasY = renderH - top * sy // flip the vertical axis
```

rdW/rdH are rounded once and reused, so the same integers drive the primary, continuous and grid tiles - a mark in one mode lands on the same pixel in every other.

How It's Calculated: Interaction

The same precision applies to what you do with the mouse.

Cursor-anchored zoom

Ctrl+wheel zooms around the cursor by a constant 10% ratio per notch: the view scales instantly while the wheel is moving and the crisp high-resolution re-render happens once when the wheel rests. The cursor position and scroll offsets are captured *before* the zoom changes, then the offsets that keep that point fixed are applied once layout settles.

```
ratio = newZoom / oldZoom
newHOff = (oldHOff + cursorX) * ratio - cursorX
newVOff = (oldVOff + cursorY) * ratio - cursorY // clamped at >= 0
```

Grid zoom by column count

Grid snaps to a whole number of columns: the count is authoritative and the zoom is derived from it, so the grid lays out exactly N pages with no leftover gap.

```
GridZoomForN(n) = (viewportW - 24) / ( n * (rdW + 12) )
```

Cache keys & eviction

The render cache is keyed by the tuple that uniquely determines a page's pixels, so a hit is always safe to reuse; whole-tab caches drop off the end of a three-deep most-recently-used list.

```
key = ( pageIndex, sizeBucket, rotation )
sizeBucket = scaledMax (primary) or round(primaryDipW) (tiles)
evict : keep the 3 most-recently-used tabs; clear the rest
```

II

Tools & Features

- Every annotation, editing and document tool
- How each behaves in the GUI
- What happens under the hood, with real names

KillerPDF

Text & Typewriter

Text `[T / 1]`

Drop a text box anywhere, or re-edit existing PDF text in place.

IN THE GUI

Click to place a box and type (Enter saves, Escape cancels); double-click existing text to re-edit. A floating bar sets color, opacity, fill, point size, font and Bold / Italic / Strike / Underline.

UNDER THE HOOD

```
PlaceTextBox builds a flat WPF TextBox with live resize handles; CommitActiveTextBox converts it to a TextAnnotation. Re-editing drops a page-color CoverAnnotation (paired by PairId) over the original; font and size are sniffed from PdfPig with a mojibake guard, falling back to Segoe UI 14pt.
```

Highlight, Strikethrough & Underline

Highlight / Strikethrough / Underline `[H / 2]`

Three text-flowing mark styles sharing one annotation type, with an optional eraser.

IN THE GUI

Drag across words and the markup hugs each line, browser-style, instead of laying down one rectangle - across lines, paragraphs and (in continuous view) across pages. One gesture makes one grouped annotation per page that selects, moves, deletes and undoes as a unit. On pages with no text layer the tools show a status hint instead of drawing a box; the highlight eraser keeps its rectangle.

UNDER THE HOOD

```
The Select tool's drag tracks the actual character runs in reading order
(Services/TextRunService.cs, PdfPig + PDFium char boxes). CommitFlowingHighlight turns the
per-line rects into HighlightAnnotations grouped by GroupId, committed as one undo step. Each is
tagged with a HighlightStyle and RGBA color.
```

Ink (Draw) & Line

Draw & Line `[D / 5 - L / 3]`

Freehand ink and straight lines, both backed by one ink annotation type.

IN THE GUI

Draw paints freehand strokes and keeps drawing without re-arming; Line drags a straight segment with an optional Level snap. The bar offers color, opacity, width and an eraser (Draw) or level (Line).

UNDER THE HOOD

> Both create an InkAnnotation (points + stroke width + RGBA, default red 2.0px). Draw accumulates points on move; Line keeps two points. A minimum-length guard rejects click jitter.

Shapes

Shapes [4]

Rectangle, ellipse and free-form polygon markers, each with an optional fill.

IN THE GUI

Box keeps the classic drag-a-filled-rectangle gesture the highlighter used to have; ellipse and polygon are closed outlines that move, resize, flatten and print like any other drawing. Free-form places points click by click - click the first point (its target lights up as you approach) or double-click to close, Esc cancels, Backspace removes the last point. The tool shares the draw bar's color, size and opacity, with a mini-shape sub-mode picker and a Fill toggle.

UNDER THE HOOD

```
New Shapes.cs partial (EditTool.Shape + ShapeKind). Box+fill is a HighlightAnnotation; box no-fill and ellipse and polygon ride the InkAnnotation pipeline with a closed flag and optional FillR/G/B/A, rendering as a WPF Polygon and exporting via gfx.DrawPolygon (fill then stroke).
```

Image & Cover

Image `[I / 6]`

Place a picture on the page; covers provide whiteout for text replacement.

IN THE GUI

Pick Image, click and choose a file (or just paste with Ctrl+V); it lands at 50% scale, corner-resizable. Covers are the opaque whiteout half of an in-place text re-edit.

UNDER THE HOOD

```
ImageAnnotation stores Base64 bytes with the decoded bitmap cached, drawn at SourceWidth*Scale.
Decoding runs through SixLabors.ImageSharp 2.1.13, and images with broken DPI metadata (WhatsApp
photos, some scans) are kept within Adobe's supported 3-14,400 point page range. CoverAnnotation
derives from HighlightAnnotation with forced-opaque alpha, paired by PairId and flattened solid on
export.
```

Crop

Crop [C / 8]

Trim pages with numeric precision, on one page or all.

IN THE GUI

A default box inset 8% from the edges plus a bar with X / Y / W / H inputs, a unit selector (points, inches or %), current- or all-pages scope, Remove-crop and Apply / Cancel.

UNDER THE HOOD

> ApplyCrop converts the canvas rect to PDF coordinates (rotation-aware via CanvasToPdfRect), clamps to the media box, writes /CropBox and /TrimBox with PdfSharpCore, then saves and reloads. Every save also strips degenerate zero-size crop boxes, healing files damaged by older builds.

Links

Links

Existing PDF links become clickable; you can copy or remove them.

IN THE GUI

Links become hand-cursor overlays: left-click follows an internal jump or external URI; right-click offers Copy URL or Remove Link. Internal jumps are recorded in the jump history, so Alt+Left retraces them. An optional Confirm before opening links setting, on the About card beside the recent-files toggle, asks before a link opens in your browser; it is off by default so links stay immediate unless you want the check.

UNDER THE HOOD

```
RenderPageLinks reads each page's /Annots, parses /Link /Rect and the /A action and builds transparent overlays; object-stream PDFs fall back to PDFium link enumeration through a cached handle that is released before every save. On save, StripLinkAnnotationBorders removes borders so they don't strobe.
```

Forms (AcroForm)

Fill, create, recover, and save interactive text fields, checkboxes, radio groups, dropdowns, list boxes, comb fields, and signature widgets.

EDITING

Field	Behavior
Text	Font sizing, alignment, Unicode text, maximum length, numeric constraints, rich metadata, and multiline input
Choice	Editable or fixed choices, selected indices, display and export values, and OCR choice-list matching
Buttons	Checkbox and radio appearance states with coordinated values and regenerated normal appearances
Comb	One character per cell with an enforced maximum length and portable appearance stream
Signature	Visual signature placement or certificate-backed CMS signing

POWERED BY THE ENGINE

The KillerPDF.Engine reads inherited field dictionaries, tracks values and defaults independently, preserves tooltips and alternate names, and writes appearance streams for reliable display in other PDF readers. Form changes participate in the same validated document save pipeline as page and annotation edits.

> The Interactive Feedback Form on page 46 includes live text, choice, button, signature, and comb examples. Saving a copy exercises the real form-writing path.

Stamp, Watermark & Page Numbers

Stamp / Watermark `[S / 0]`

Two independent overlays - page numbers and a text or image watermark.

IN THE GUI

A live-preview editor with a page stepper. Page numbers take a start value, a format ({n} current, {N} total), size, color, a range like 1-3,5, eight positions plus custom and a spread mirror. Watermarks take text or an image, size / color or scale, opacity, angle and position. Stamps apply to a clean document with no other markup, and clear again by applying with both sections off.

UNDER THE HOOD

```
A StampSpec holds every setting (watermark default DRAFT 64pt 25% at 45 degrees).
DrawStampsOnDocument flattens via PdfSharpCore XGraphics - watermark under, numbers on top, image
opacity baked through a color matrix, rotation via translate-plus-rotate.
```

Search

Search `[Ctrl+F - F3]`

Debounced full-text search with persistent per-page highlights.

IN THE GUI

Ctrl+F toggles a draggable search bar (250ms debounce); the status shows current / total, and F3 / Shift+F3 step to the next / previous match from anywhere (F3 opens search when it isn't showing). Enter, Shift+Enter and the wheel walk the matches too.

UNDER THE HOOD

```
SearchService.Search scans each page's PdfPig GetWords() case-insensitively (single- and
multi-word) for per-page rectangles. ApplySearchHighlights repaints them on every render - the
current match a brighter accent outline, the rest dim.
```

OCR

OCR `[Ctrl+Shift+O / Ctrl+Shift+I]`

On-device text recognition in nineteen languages with searchable-PDF output and form-field recovery.

IN THE GUI

Ctrl+Shift+O recognizes the current page to the clipboard. Ctrl+Shift+I drags a region. The OCR menu manages installed models and can write transparent recognized text back over a scan so it becomes selectable and searchable without changing its appearance.

LANGUAGE CATALOG

English, Bengali, Czech, German, Spanish, French, Hungarian, Italian, Japanese, Kazakh, Norwegian, Polish, Portuguese, Russian, Turkish, Ukrainian, Vietnamese, Chinese Simplified, and Chinese Traditional. Standard and high-accuracy model tiers download on demand.

NEW FORM RECOVERY

OCR can rebuild field geometry and preserve practical constraints such as numeric input, maximum lengths, comb cells, and choice-list matching. Recovered widgets use Engine-generated appearances that remain readable outside KillerPDF.

UNDER THE HOOD

> `OcrNativeBootstrap extracts the x64 Tesseract and Leptonica runtime to a per-version cache. Pages render at about 300 DPI and feed OcrService, which returns text and confidence while all recognition stays on the device.`

Digital & Visual Signing

DIGITAL SIGNATURES

Create certificate-backed CMS SHA-256 signatures from PFX or P12 files, the Windows certificate store, and compatible USB security tokens. Add reason, location, contact information, and an editable visible signature layout before writing the signed copy.

Stage	Protection
Certificate	Valid signing certificate and private-key availability are checked before output
Byte ranges	Reserved signature contents and covered byte ranges are calculated by the Engine
Visible widget	Text, certificate details, image, placement, and appearance are editable
Later edits	A stale cryptographic value is removed rather than presented as a valid signature

VISUAL SIGNATURES

Draw a signature at three pen widths or import one, save signatures and initials for reuse, and drag them onto a page or compatible form field. Visual signatures are annotations and do not claim cryptographic identity.

> Digital signing proves document integrity and certificate identity. A visual signature is an editable mark. KillerPDF labels and stores the two workflows separately.

Print

Print `[Ctrl+P]`

A live print-preview window with printer-aware paper, layout, output, and progress controls.

PRINTER

Choose the printer, paper size, and paper source. Match document automatically selects an appropriate sheet when the driver exposes one.

LAYOUT

Portrait or landscape, nine placement anchors, none-to-wide margins, 1, 2, 4, 6, or 9 pages per sheet, and fit, actual-size, or custom scaling. PRINTER, LAYOUT, and OUTPUT are collapsible sections.

OUTPUT

Color or black-and-white, copies, page ranges, odd/even subsets, and printer-supported two-sided output. Preferences for printer, orientation, color, and duplex persist.

PREVIEW AND SPOOL

Pages render progressively with a live count, navigation remains available, and Print stays disabled until the preview is ready. Final 300 DPI preparation and asynchronous spooling show progress instead of freezing the window.

> The preview and final job use the same sheet composer, so scale, margins, alignment, page subsets, N-up placement, copies, and duplex padding match what is shown.

Rotate & Transform

Rotate / Transform [R / 9]

Use Transform when pages are sideways, stretched, mirrored, tilted, noisy, or too pale to read. Version 1.8.0 adds selected-page batches, grayscale and threshold output, 72 to 600 DPI rendering, and JPEG compression controls.

1. **OPEN** — R or toolbar slot 9
2. **FIX SHAPE** — rotate, scale, flip, level
3. **FIX TONE** — black, white, midtones
4. **APPLY** — one page or a batch

WHICH CONTROL SHOULD I USE?

Problem	Control	What to do
Sideways or upside down	Rotate	Choose 90 degrees, or enter any angle. Expand keeps corners from being clipped.
Page is too large or small	Scale	Change width and height together, or unlock them to correct stretching.
Text reads backward	Flip	Mirror horizontally; vertical flip handles an upside-down scan without rotating.
Lines climb or fall	Skew / Level	Enter a small angle, or draw the Level line along text that should be horizontal.
Gray paper, pale text	Levels	Tune black, white, and midtones, then choose color, grayscale, or threshold output.

A CLEAN TRANSFORM WORKFLOW

Order	Why
1. Rotate or level	Establish the page's true horizontal and vertical axes first.
2. Scale or flip	Correct dimensions and mirroring after the page is oriented.
3. Adjust Levels	Tune black point, white point, and midtones after geometry is settled.
4. Apply once	Commit the preview as one undoable page replacement.

WHAT STAYS SAFE

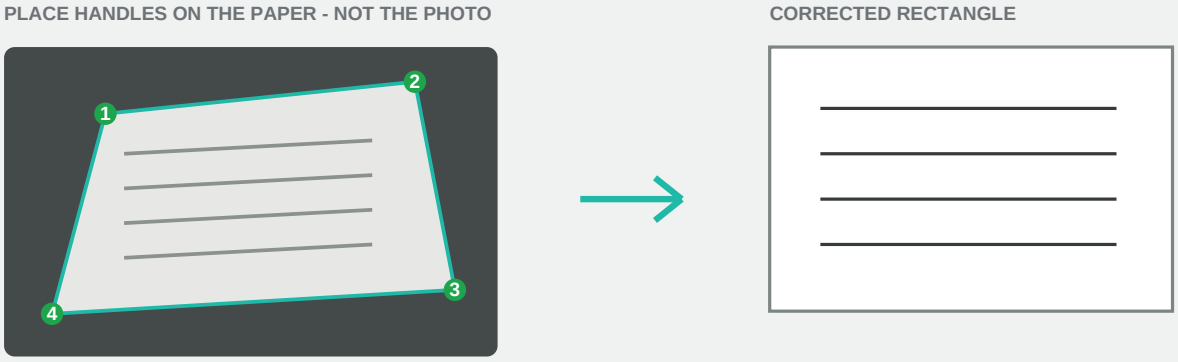
The full-page preview is non-destructive. Reset inside one section clears only that section, Reset all returns to the original, and closing the window without Apply changes nothing. Page Quality offers color, grayscale, or black-and-white output, a visible 0 to 255 threshold, optional 72 to 600 DPI output, and optional JPEG compression from 25% to 100%.

> Apply rebuilds each selected page once, moves its annotations with it, and records the batch in document undo. Black-and-white pages save as compact 1-bit images without retaining the old color resources. Ctrl+Z restores the original pages and annotation geometry.

Perspective Correction

Straighten a photographed page [R / 9]

Use Perspective when a rectangular page, sign, screen, or whiteboard looks narrower on one side because the camera was not directly in front of it. Do not use it for a simple tilted scan - the Level line is faster for that.



DO THIS

Step	Action
1. Turn it on	Open Transform, expand PERSPECTIVE, and enable Correct trapezoidal distortion.
2. Place corners	Drag the handles in order: top-left, top-right, bottom-right, bottom-left. Match the real paper corners.
3. Check the preview	The page should become rectangular. If it twists or folds, two handles crossed or landed on the wrong edge.
4. Apply	Apply rebuilds the page at full resolution. Ctrl+Z restores the original if the result is not right.

BEST RESULTS AND COMMON MISTAKES

If you see...	Try this
A pinched or twisted result	Keep the handles clockwise and make sure their connecting sides never cross.
A border still leaning	Zoom in and put each handle on the paper edge itself, not its shadow or the photo boundary.
A page that is only tilted	Turn Perspective back off and draw the Level line along a line of text instead.
Pale text after correction	Finish the geometry first, then open LEVELS and adjust black point, white point, and midtones.

> Under the hood: the four normalized corners form a validated, non-crossing quadrilateral. KillerPDF maps that shape to a rectangle with a projective homography, samples it at full transform resolution, and composes the correction with rotation, skew, scale, flip, and Levels as one undoable page edit.

Page Operations & Import

Build and reorganize documents directly from the Pages panel, the split workspace, Windows Explorer, and the command line.

Operation	1.8.0 behavior
Multi-page drag	Selected thumbnails move as one ordered block and keep their relative order
Exact-position drop	PDFs and images insert at the visible page insertion line, not only at the end
Cross-pane transfer	Drag selected pages between documents in Split Pane
Import	PDF pages, images, folders, and zip contents become validated document pages
Duplicate / delete	Batch page actions participate in document undo and preserve selection
Export	Selected pages or page images can be written without changing the source document

ENGINE-BACKED INSERTION

Imported pages are copied through The KillerPDF.Engine object graph. Resources, inherited page settings, annotations, forms, links, metadata references, and content streams are transferred with validated ownership instead of raw object-number reuse.

> The insertion line follows the active theme accent. A multi-selection moves atomically, so a cancelled or invalid transfer does not leave a partially edited document.

Split Pane

Split Pane [F10]

Two complete document workspaces side by side, with page transfer and synchronized comparison.

IN THE GUI

F10 opens or closes the second pane. Focus decides which pane the shared toolbar, footer view and zoom controls, sidebar, page list, dialogs, and commands act on. Tabs and selected pages can be dragged across the divider, and each side resizes without collapsing below a readable width.

INDEPENDENT PANE STATE

Per pane	Independent behavior
Documents and tabs	Each side owns its sessions, active tab, and deferred opens.
View	Current page, zoom, mode, scroll position, and render cache.
Editing	Tool, selection, annotations, forms, links, crop, and text editing.
Night mode	N flips only the focused pane; the moon icon follows focus.

COMPARE PDFS

Comparison opens with each document fitted to its pane's width, pairs pages, and synchronizes navigation and scrolling. Fit modes adapt to each pane; an explicit zoom percentage applies to both. Difference regions identify changes, dimension mismatches, and pages present in only one document. The original files remain untouched.

UNDER THE HOOD

> The reusable PdfViewer owns document interaction and rendering. MainWindow owns shared application chrome and routes each command through ActiveViewer, preventing state from leaking across panes.

Tabs, Outline & Thumbnails

Tabs `[Ctrl+Tab - Ctrl+W - Ctrl+Shift+W]`

Multi-document tabs with flicker-free, reconcile-in-place rendering.

IN THE GUI

Open many documents as tabs; the active tab stays visible with an overflow menu. Ctrl+Shift+W (or Close Other Tabs on a tab's right-click menu) closes every tab except the current one. Collapsing the sidebar is a smooth quarter-second slide.

UNDER THE HOOD

```
Backed by DocumentSession. The strip is reconciled in place via a session-to-element cache - no
re-parenting, so no flicker on switch, drag or resize. Lazy tabs defer loading until activated.
```

Outline & Thumbnails `[Ctrl+B]`

A dockable sidebar with an editable bookmark tree and lazy thumbnails.

IN THE GUI

A dockable sidebar carries a Pages tab and an Outline tab that mirrors the PDF bookmark tree. A compact header control expands or collapses every branch. Bookmarks can also be added, renamed inline (F2), nested, reordered, retargeted, deleted, and multi-selected with full undo.

UNDER THE HOOD

```
LoadOutlines walks _doc.Outlines into a TreeView; titles from password-protected files are
re-decoded to dodge mojibake, and bookmarks that point at named destinations resolve through the
catalog name-tree. Thumbnails lazy-load on a concurrency-limited thread at ~288px.
```

Redo & Jump History

Redo / Jump history `[Ctrl+Y - Alt+Left / Alt+Right]`

Browser-style history for both your edits and your movements through the document.

IN THE GUI

Ctrl+Y (or Ctrl+Shift+Z) re-applies undone actions: annotations, text edits, stamps, clears and document-level operations alike. Alt+Left / Alt+Right, or the mouse back / forward buttons, retrace bookmark, link, jump-box and Home / End jumps exactly like a browser.

UNDER THE HOOD

```
Redo entries mirror the _undoStack records; any new edit clears the redo chain, and both stacks
live per tab in the DocumentSession. Jump history is a pair of page stacks (back / forward) fed by
RecordNavJump at every hard jump; retracing pushes the current page onto the opposite stack.
```

Zoom & Undo

ZOOM

Cursor-anchored zoom from 5% to 500%.

IN THE GUI

The footer groups zoom out, the editable zoom box, and zoom in beside the four view-mode buttons. Enter a percentage or select Fit Width or Fit Page. Click a pane to make it the target of these controls.

Ctrl+wheel zooms around the cursor; in Grid it steps the column count. Ctrl+plus and Ctrl+minus zoom in and out. Ctrl+1, Ctrl+2, and Ctrl+3 select actual size, fit width, and fit page.

UNDER THE HOOD

```
ZoomMin 0.05, ZoomMax 5.0, ZoomStep 0.15; FitMode is None / Width / Page. See the rendering and interaction pages for the cursor-anchor math.
```

UNDO [CTRL+Z]

One history covers annotations, pages, and whole-document edits. Ctrl+Z reverses the last change; auto-repeat is ignored so one keypress is one undo.

UNDER THE HOOD

```
The undo stack holds records for individual and grouped annotations, page clearing, stamp batches, bookmark edits, page snapshots, and full-document byte snapshots.
```

Command Line

Every core operation also runs headless from a terminal. No window, meaningful exit codes (0 success, 1 failed, 2 bad usage), and it works even while the app is open. Each command reuses the exact pipeline its GUI equivalent runs.

```
KillerPDF.exe --merge out.pdf a.pdf b.pdf scan.jpg
KillerPDF.exe --extract-pages in.pdf 1-3,5 out.pdf
KillerPDF.exe --split in.pdf pages\
KillerPDF.exe --decrypt locked.pdf open.pdf [--password p]
KillerPDF.exe --to-image in.pdf imgs\ --dpi 300 --format jpg
KillerPDF.exe --flatten in.pdf flat.pdf
KillerPDF.exe --print in.pdf --printer "HP LaserJet" --pages 1-4
KillerPDF.exe --ocr scan.pdf searchable.pdf --lang eng
KillerPDF.exe --batch-resave inDir\ outDir\ --log report.csv
KillerPDF.exe --help
```

Batch mode & the validation harness

--batch-resave resaves a single PDF or a whole folder tree headlessly through the standard open/save pipeline with per-file OK / SKIP / FAIL reporting. It exists so the standards-conformance harness (validation/, see Part III) can push thousands of corpus files through a real KillerPDF save and prove the output validates exactly as well as the input.

Rasterizing is rotation-safe at 150 or 300 DPI, printing drives the same 300 DPI flattened pipeline as the dialog, and --version / --help print and exit cleanly for scripting.

Reference

- The complete keyboard map and tool keys
- Thirteen themes and eight accent colors
- Nineteen languages, standards, and measured results

Keyboard Shortcuts

The F1 shortcut overlay shows the same bindings in LIST and KEYBOARD views. Global tool keys pause while typing in a text or form field.

FILE

Key	Action
Ctrl+O	Open document
Ctrl+S	Save
Ctrl+Shift+S	Save As
Ctrl+W	Close file
Ctrl+Shift+W	Close other files
Ctrl+Q	Close all files
Ctrl+N	New blank document
Ctrl+P	Print Preview
Ctrl+D / F4	Document info
Shift+F4	Current file size

TOOLS

Key	Action
V	Select
1 / T	Text
2 / H	Highlight
3 / L / U	Line
4	Shapes
5 / D	Draw
6 / I	Image
7 / G	Signature
8 / C	Crop
9 / R	Transform
Ctrl+R	Rotate clockwise
Ctrl+Shift+R	Rotate counterclockwise
0 / S	Stamp pages
F	Form field

Keyboard Shortcuts (continued)

EDIT

Key	Action
Ctrl+Z	Undo
Ctrl+Y	Redo
Ctrl+Shift+Z	Redo
Ctrl+C	Copy text
Ctrl+V	Paste
Ctrl+B / I / U	Bold / italic / underline
Delete	Delete annotation or bookmark
F2	Rename bookmark
Enter / Esc	Confirm / cancel
Menu / Shift+F10	Context menu

HELP

Key	Action
F1 / Ctrl+?	Shortcut overlay
F12	About
Ctrl+Shift+K	Toggle shortcut hints

NAVIGATION

Key	Action
Left / Right / PgUp / PgDn	Previous / next page
Home / End	First / last page
Alt+Left / Alt+Right	Back / forward history
Up / Down	Scroll view
Ctrl+Scroll	Zoom at cursor
Ctrl++ / Ctrl+-	Zoom in / out
Ctrl+0	Reset zoom
Ctrl+1 / 2 / 3	Actual / fit width / fit page
Middle-drag	Pan view
Space-drag	Pan view
F9	Toggle sidebar
Shift+F9	Move sidebar
Ctrl+Tab	Next tab
Ctrl+Shift+Tab	Previous tab

Keyboard Shortcuts (continued)

VIEW

Key	Action
F5	Continuous
F6	Single page
F7	Two-Page
B	Book layout
F8	Grid
F10	Split pane
F11 / Esc	Full screen
Alt+M	Hide toolbar
N	Invert focused pane
Shift+N	Also invert images
Ctrl+Shift++ / - / 0	Application size
Wheel on logo	Application size
Ctrl+Shift+1-6	Toolbar icon and label layout

OCR

Key	Action
Ctrl+Shift+O	OCR current page
Ctrl+Shift+I	OCR region

SEARCH AND SELECTION

Key	Action
Ctrl+F	Find
F3 / Shift+F3	Next / previous result
Enter / Shift+Enter	Next / previous result
Ctrl+A	Select all
Shift+Click	Multi-select

Tool Keys

Digits 1-9 and 0 follow toolbar order. The letter keys are quick aliases. Press Esc with no active tool to return to Select.

Key	Action
V	Select
T / 1	Text
H / 2	Highlight
L / U / 3	Line
4	Shapes
D / 5	Draw
I / 6	Image
G / 7	Signature
C / 8	Crop
R / 9	Transform
S / 0	Stamp
F	Form field

REFERENCE

Themes & Accents

Thirteen built-in themes share one semantic resource map. Gradient themes keep their full outer-window color in About and Keyboard Shortcuts overlays.

Dark `#1EA54C`
Charcoal and brand green

Light `#18843D`
Light surfaces and deep green

Black `#00FF66`
High contrast neon chrome

Blood `#F8C99E`
Oxblood and cream

Greed `#E0D49A`
Forest green and gold

Cyanotic `#E0D49A`
Deep blue and warm accent

98SE `#004F00`
Classic beveled Windows chrome

Ectoplasm `#EAD900`
Purple-to-charcoal gradient

Decay `#B8AA7C`
Slate-to-olive gradient

Mourning `#FF6F91`
Muted violet and pink

Sepulchre `#4FAAA8`
Charcoal-to-teal gradient

Delirium `#DD8500`
Oxblood-to-violet gradient

Malaise `#FF6F91`
Mauve-to-teal gradient

Accent variants

Dark, Light, and Black offer eight accents:

Red | Orange | Yellow | Green | Teal | Blue | Purple | Magenta

Localization

KillerPDF ships nineteen interface languages. English remains the base dictionary, so a missing translated key falls back safely while theme and language switches repaint live.

Locale	Name
English (en-US)	English
Spanish (es)	Español
German (de-DE)	Deutsch
French (fr-FR)	Francais
Turkish (tr-TR)	Turkce
Bengali (bn)	Bangla
Japanese (ja-JP)	Nihongo
Chinese Simplified (zh-CN)	Simplified
Chinese Traditional (zh-TW)	Traditional
Czech (cs-CZ)	Cestina
Polish (pl-PL)	Polski
Hungarian (hu-HU)	Magyar
Italian (it-IT)	Italiano - new in 1.8.0
Kazakh (kk-KZ)	Qazaqsha - new in 1.8.0
Russian (ru-RU)	Russkiy
Vietnamese (vi-VN)	Vietnamese
Norwegian Bokmål (nb-NO)	Norwegian Bokmål
Portuguese (Brazil) (pt-BR)	Brazilian Portuguese
Ukrainian (uk-UA)	Ukrainian

LocaleManager merges theme + en-US + the chosen locale. Code-built context menus, toolbar captions, annotate bars, dialogs, and the shortcuts overlay rebuild when the language changes. Development installs can load and live-reload an external translation file.

REFERENCE

File Handling & Robustness

KillerPDF is designed to open difficult PDFs, preserve recoverable content, and stop before an unsafe operation can damage the user's only copy.

RESILIENT OPEN

- Network and UNC files are staged locally when needed to avoid short-read behavior, while encrypted documents request credentials without exposing them in logs.
- Malformed cross-reference data, page trees, stream lengths, inherited resources, and object streams are normalized into an Engine document model with explicit diagnostics.
- Unsupported structures can open read-only or produce a safe error. The original file is never rewritten merely because it was opened.

VALIDATED SAVE

Stage	Behavior
Prepare	Apply page, annotation, form, metadata, encryption, and signature changes to an isolated document model
Validate	Check references, ownership, page structure, streams, appearance state, and output constraints
Write	Use an incremental update or deterministic full rewrite through The KillerPDF.Engine
Commit	Replace the requested output only after the complete file is written and closed successfully
Reload	Reopen the saved result and refresh sessions without duplicating flattened marks

RECOVERY AND REPORTING

Document undo records page and annotation operations before save. Crash reports include a bounded timestamped status history and rotate at a fixed size. Batch tools return per-file OK, SKIP, or FAIL results so a large job can continue without hiding individual failures.

> The compact installer and portable package use verified payloads. Installation-scope guards prevent a per-user and all-users copy from silently competing on the same computer.

Standards Conformance

The KillerPDF.Engine targets modern PDF syntax and validates release output against both normative structure rules and real-world hostile documents.

SUPPORTED DOCUMENT FAMILIES

Family	Coverage
PDF	Classic and cross-reference streams, object streams, incremental updates, encryption, signatures, forms, annotations, outlines, and tagged structure
PDF 2.0	Modern object and metadata rules, Unicode text, current signature structures, and PDF 2.0 document output
PDF/A	PDF/A-1, PDF/A-2, and modern PDF/A-4 validation paths, including metadata and transparency constraints
PDF/UA	Tagged structure preservation and PDF/UA-1 and PDF/UA-2 validation inputs

RELEASE GATE

The 47,024 inputs include 16,696 public regression files, 649 standards/color files (552 public and 97 local), 29,599 private stress files, and 80 damaged inputs. Each main collection had one warmup and five measured runs; damaged inputs were tested individually.

1.8.3 BATCH OPEN/SAVE RESULTS

Check	Result
Corpus coverage	47,024 inputs across four suites
Repeated runs	Five matching measured runs per main collection
Additional saves	1,485 more than 1.8.2
Previously successful saves lost	0

> These results measure batch opening and saving, not rendering accuracy or PDF conformance. This run did not include a new qpdf/veraPDF conformance sweep. Page 42 records the counts and timings.

REFERENCE

Engine Validation & Guarantees

The KillerPDF.Engine is tested as a document library, not only through the desktop interface. Unit, integration, round-trip, corpus, qpdf, and veraPDF checks cover both ordinary documents and deliberately hostile inputs.

RELEASE EVIDENCE

Measure	1.8.3 result (September 2, 2026)
Engine tests	1,443 passing
Application tests	289 passing
Corpus inputs	47,024 PDFs across four suites
Regression suite	15,171 saved, 1,233 skipped, 292 save failures
Standards suite	377 saved, 235 skipped, 37 save failures
Stress suite	25,015 saved, 2,957 skipped, 1,627 save failures
Malformed suite	78 skipped, 2 save failures; no crashes or timeouts

MEASURED TIMING AND STABILITY

Suite	Files	Median time / repeatability
Regression	16,696	182.093 s; five matching runs
Standards/color	649	5.666 s; five matching runs
Stress	29,599	523.400 s; five matching runs
Malformed	80	Individual 30-second limits; no timeouts

FAIL CLOSED

Before writing, the Engine validates references, page trees, stream lengths, encryption state, signature reservations, form relationships, appearance state, and output constraints. Unsupported unsafe edits become explicit errors or safe skips. They do not become silent file damage.

> `40,563 files saved, 1,485 more than 1.8.2. Every previously successful file still saved. Regression took 8.5% longer, stress 2.0% longer, and standards/color was essentially unchanged. Full report: KillerPDF.net/corpus.html.`

REFERENCE

Specifications & Constants

Selected limits and defaults drawn from the 1.8 source.

Specification	Value
Render coordinate space	2048 DIP longest side
Bitmap cap	6144 px
Print / OCR render	300 DPI on demand / about 300 DPI
Image export	24-1200 DPI, PNG or JPEG
Zoom range	5% to 500%
App size	70% to 250%
Page bitmap cache	About 160 MB per tab, budgeted in bytes
Per-file state	40 documents
Folder / zip import	50 files per drop
Signature contents	16,384 bytes, CMS SHA-256
Default text	Segoe UI, 14 pt
Crash logs	50 status entries; 20 MB rotation
UI languages / themes	19 locales / 13 themes / 8 neutral accents
Engine tests (1.8.3)	1,443 passed; 0 failed; 0 skipped
Application tests (1.8.3)	289 passed; 0 failed; 0 skipped

Limits and defaults describe KillerPDF 1.8. All 1,732 engine and application tests passed on September 2, 2026.

Glossary

Plain-language definitions for the terms used throughout this brochure.

Term	What it means
WPF	Windows Presentation Foundation, Microsoft's native desktop UI framework
partial class	A C# feature that lets one class be written across several files
PDFium	Google's open-source PDF rendering engine that turns a page into pixels
Docnet.Core	The .NET wrapper KillerPDF uses to drive PDFium
PdfPig	A .NET library that extracts text and word positions from a PDF
PdfSharpCore / PDFsharp	Libraries that write PDFs: saves and edits (vendored and patched), and signing
vendored	A library's source copied into the repo and built with the app, so it can be patched and audited
veraPDF	The industry reference validator for the PDF/A and PDF/UA standards
PDF/A	The ISO standard for archival PDFs; strict rules so files stay readable for decades
Tesseract	The open-source OCR engine that recognizes text in images
DIP	Device-independent pixel, a unit that stays the same apparent size on any display
render-dim	The internal page size (longest side 2048) where annotations are stored
xref	The cross-reference table, a PDF's index of where each object lives
MediaBox	The PDF entry that defines a page's full size
AcroForm	The PDF standard for interactive, fillable form fields
CMS / PKCS#7	The standard cryptographic container that holds a digital signature
OCR / DPI	Optical character recognition / dots per inch (300 DPI is print quality)

IV

Live Showcase

- A real interactive form to fill and save
- A printed-style form to mark up by hand
- Annotation pages seeded for screenshots

KillerPDF

LIVE SHOWCASE

Interactive Feedback Form

Real AcroForm fields - open in KillerPDF (or any reader), fill them in, then Save.

Name

Role

Email

Version used (comb)

Which tools do you use most? (tick all that apply)

- ☐ Annotation
- ☐ Forms
- ☐ OCR
- ☐ Signing
- ☐ Shapes
- ☐ Crop / pages
- ☐ Search
- ☐ Print

Overall rating

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5 (best)
- ☐ Notify me of releases

Comments

Signature

Click the signature field in KillerPDF to sign here.

Date

Initials

FORM FEATURES ON THIS PAGE

Text | dropdown | comb | checkbox | radio | multiline | signature | initials

Focus text to adjust its size. Save keeps the fields interactive.

Sample Application Form

A printed-style form with no interactive fields - ideal for demoing fill-by-annotation, crop and stamp.

KILLERPDF - USER REGISTRATION & EVALUATION

Form KP-1 (Rev. 1.8) - Please print clearly in black ink

LAST NAME _____ FIRST NAME _____

ORGANIZATION _____ EMPLOYEE ID _____

EMAIL ADDRESS _____

PHONE _____ DATE (YYYY-MM-DD) _____ LICENSE No. _____

Edition requested: ☐ Portable (.exe) ☐ Installer ☐ Source build (GPL-3) ☐ Enterprise / MSP

Platform: ☐ Windows 10 ☐ Windows 11 ☐ Server 2019+ ☐ Other _____

Comments

SIGNATURE OF APPLICANT _____ DATE _____

LIVE SHOWCASE

Annotation Playground

Seeded content for screenshots. Reach for the toolbar (or the number keys) and mark this page up.

Highlight me (H / 2)

Drag the Highlight tool across this sentence and the mark flows along the words; Strikethrough and Underline give the other two styles.

Draw a shape (4)

Reach for the Shapes tool and drop a rectangle, ellipse or free-form polygon over this row, with an optional fill.

Draw here (D / 5)

Freehand ink area - round caps, adjustable width, optional eraser. The minimum-length guard ignores accidental taps.

Type a note (T / 1)

Click with the Text tool to drop a typewriter note, then style it from the floating bar.

Mark up a table

Tool	Shortcut	Output type
Highlight	H / 2	Highlight annotation
Shapes	4	Ink / highlight (filled)
Draw	D / 5	Ink annotation
Stamp	S / 0	Flattened overlay
Crop	C / 8	CropBox / TrimBox

LIVE SHOWCASE

Annotation Playground II

Seeded content for screenshots. Reach for the toolbar (or the number keys) and mark this page up.

Highlight me (H / 2)

Drag the Highlight tool across this sentence and the mark flows along the words; Strikethrough and Underline give the other two styles.

Draw a shape (4)

Reach for the Shapes tool and drop a rectangle, ellipse or free-form polygon over this row, with an optional fill.

Draw here (D / 5)

Freehand ink area - round caps, adjustable width, optional eraser. The minimum-length guard ignores accidental taps.

Type a note (T / 1)

Click with the Text tool to drop a typewriter note, then style it from the floating bar.

Mark up a table

Tool	Shortcut	Output type
Highlight	H / 2	Highlight annotation
Shapes	4	Ink / highlight (filled)
Draw	D / 5	Ink annotation
Stamp	S / 0	Flattened overlay
Crop	C / 8	CropBox / TrimBox



KillerPDF

Colophon

DESIGN & DEVELOPMENT

KillerPDF is designed and built by Steve the Killer.

ICON

Designed and drawn by Steve the Killer in GIMP.

BROCHURE

Updated for v1.8; features and constants are drawn from the application source.

LICENSE

KillerPDF is free software released under GPL-3.0.

Body type is Arial. Technical callouts use Courier. Titles use Killer Sans; the wordmark pairs it with Freckle Face.

KillerPDF.net